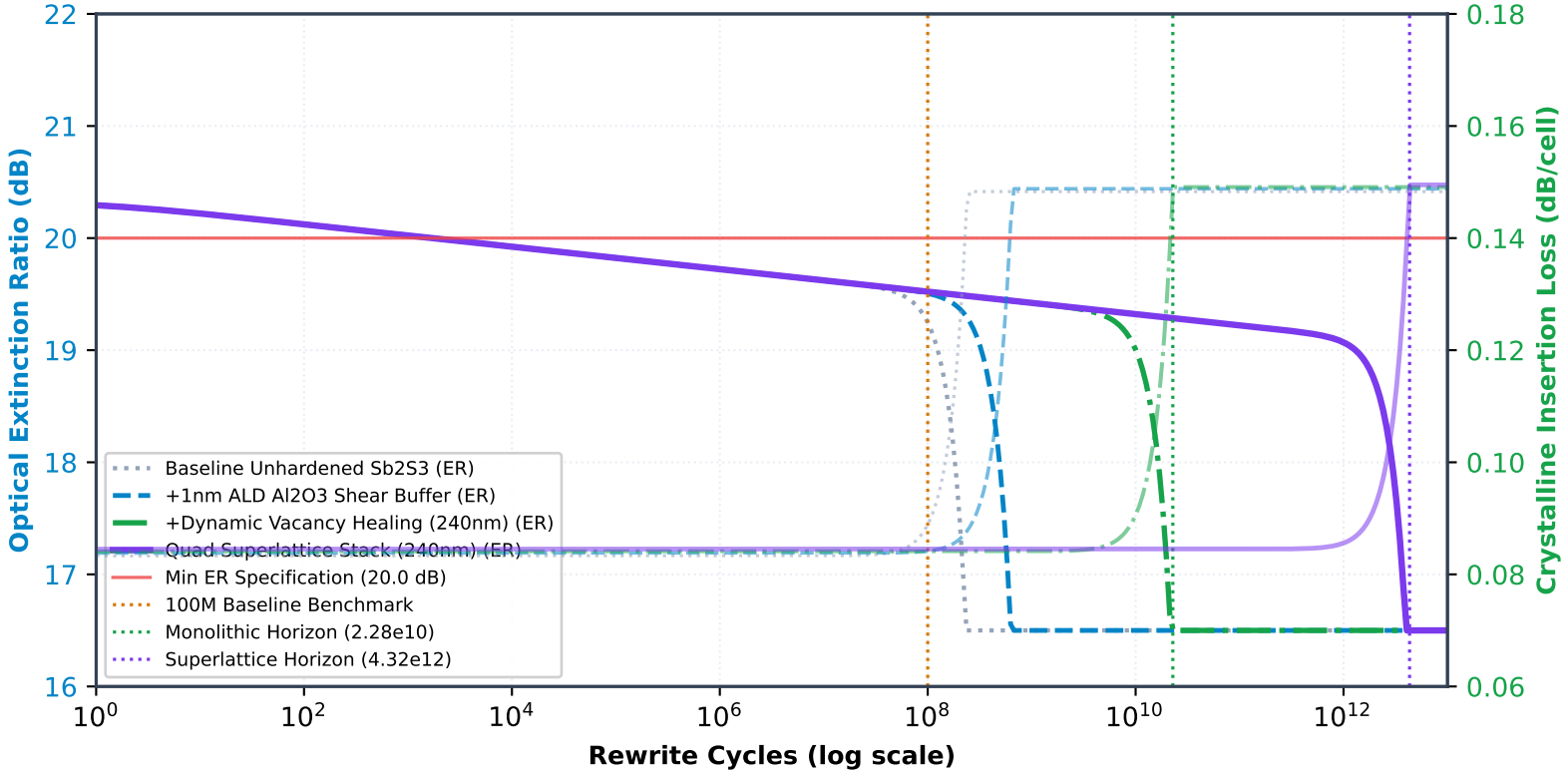
Sb_2S_3 Hardened Switch: 10^8 to 4.32×10^{12} Rewrite Cycle Retention
Optical Extinction Ratio (dB)
Crystalline Insertion Loss (dB/cell)
Rewrite Cycles (log scale)
22
21
20
19
18
17
16
0.18
0.16
0.14
0.12
0.10
0.08
0.06
10^0
10^2
10^4
10^6
10^8
10^{10}
10^{12}
Baseline Unhardened Sb2S3 (ER)
+1nm ALD Al2O3 Shear Buffer (ER)
+Dynamic Vacancy Healing (240nm) (ER)
Quad Superlattice Stack (240nm) (ER)
Min ER Specification (20.0 dB)
100M Baseline Benchmark
Monolithic Horizon (2.28e10)
Superlattice Horizon (4.32e12)